Then, it is easy to get

$$\tan(-\theta) = -\tan\theta,\ \text{cosec}(-\theta) = -\text{cosec}\,\theta,\ \sec(-\theta) = \sec\theta,\ \cot(-\theta) = -\cot\theta.$$

(i) $\sin(-\theta) = -\sin\theta$, and $\cos(-\theta) = \cos\theta$. These facts follow from the symmetry of the unit circle about the x-axis. The angle $-\theta$ is the same as angle θ except it is on the other side of the x- axis. Flipping a point (x, y) to the other side of the x- axis makes the point into $(x, -y)$, so the y-coordinate is negated and hence the sine is negated, but the x-coordinate remains the same and therefore the cosine is unchanged.

(ii) The negative-angle identities can be used to determine if a trigonometric function is an odd function or an even function .

Example 3.10 Find the values of (i) $\sin(-45^\circ)$ (ii) $\cos(-45^\circ)$ (iii) $\cot(-45^\circ)$

Solution:

(i) $\sin(-45^\circ) = -\sin(45^\circ) = -\dfrac{1}{\sqrt{2}}$.

(ii) $\cos(-45^\circ) = \dfrac{1}{\sqrt{2}}$ and $(iii)\ \cot(-45^\circ) = -1$

We have already learnt the trigonometric ratios of the angle $(90^\circ - \theta)$, $\left(0 < \theta < \dfrac{\pi}{2}\right)$ in the lower class. Let us recall the trigonometric ratios of angle $(90^\circ - \theta)$;

$$\begin{aligned} \sin(90^\circ - \theta) &= \cos\theta, \quad \cos(90^\circ - \theta) = \sin\theta, \quad \tan(90^\circ - \theta) = \cot\theta \\ \text{cosec}(90^\circ - \theta) &= \sec\theta, \quad \sec(90^\circ - \theta) = \text{cosec}\,\theta, \quad \cot(90^\circ - \theta) = \tan\theta. \end{aligned}$$

Now, we will establish the corresponding trigonometric ratios for an angle of the form $(90^\circ + \theta)$.

Trigonometric ratios of an angle of the form $(90^\circ + \theta)$, $\left(0 < \theta < \dfrac{\pi}{2}\right)$ in terms of θ .

Let $\angle AOL = \theta$ and $\angle AOR = (90^\circ + \theta)$. Let $P(a, b)$ be a point on OL and choose a point P' on OR such that $OP = OP'$.

Draw perpendiculars PM and $P'N$ from P and P' on Ox and Ox' respectively.

Now, $\angle AOP' = 90^\circ + \theta$.

Clearly, $\triangle OPM$ and $\triangle P'ON$ are congruent.

$$ON = MP \text{ and } NP' = OM$$

Hence, the coordinates of P and P' are $P(a, b)$ and $P'(-b, a)$, respectively. Now

$$\sin(90^\circ + \theta) = \frac{y\text{-coordinate of } P'}{OP'} = \frac{a}{OP} = \cos\theta,$$

$$\cos(90^\circ + \theta) = \frac{x\text{-coordinate of } P'}{OP'} = \frac{-b}{OP} = -\sin\theta,$$

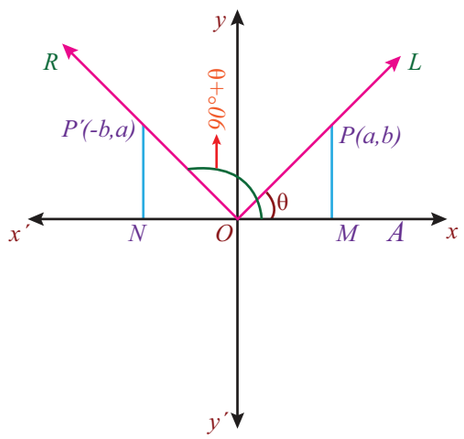


Figure 3.12

Thus, $\tan(90^\circ+\theta) = -\cot\theta$, $\text{cosec}(90^\circ+\theta) = \sec\theta$, $\sec(90^\circ+\theta) = -\text{cosec}\theta$, $\cot(90^\circ+\theta) = -\tan\theta$.

The trigonometric function of other allied angles $\pi \pm \theta$, $\dfrac{3\pi}{2} \pm \theta$, $2\pi \pm \theta$ can be obtained in a similar way.

The above results can be summarized in the following table: $\left(\text{Here } 0 < \theta < \dfrac{\pi}{2}\right)$